

INTRODUCTION

A single stray error could corrupt the information stored in a quantum computer. Quantum error correction seeks to solve this by encoding one logical qubit into several physical qubits.¹ The Steane code uses seven physical qubits and can correct any single-qubit error. In theory, it relies on errors arriving one data qubit at a time. Yet, in practice, the process of checking code introduces its own vulnerabilities.² There are many well-developed gadgets to account for these. One proposed solution is a flag-qubit circuit: an extra ancilla that monitors the syndrome ancilla during the most susceptible part of the sequence, and “flags” when a fault occurs.

In this project, we ask whether the addition of the flag qubit gadget to the X-stabilizer measurement circuit of the Steane code reduces the rate of undetected, uncorrectable errors caused by fault propagation. If so, what becomes the new qubit and gate overhead?

METHODS

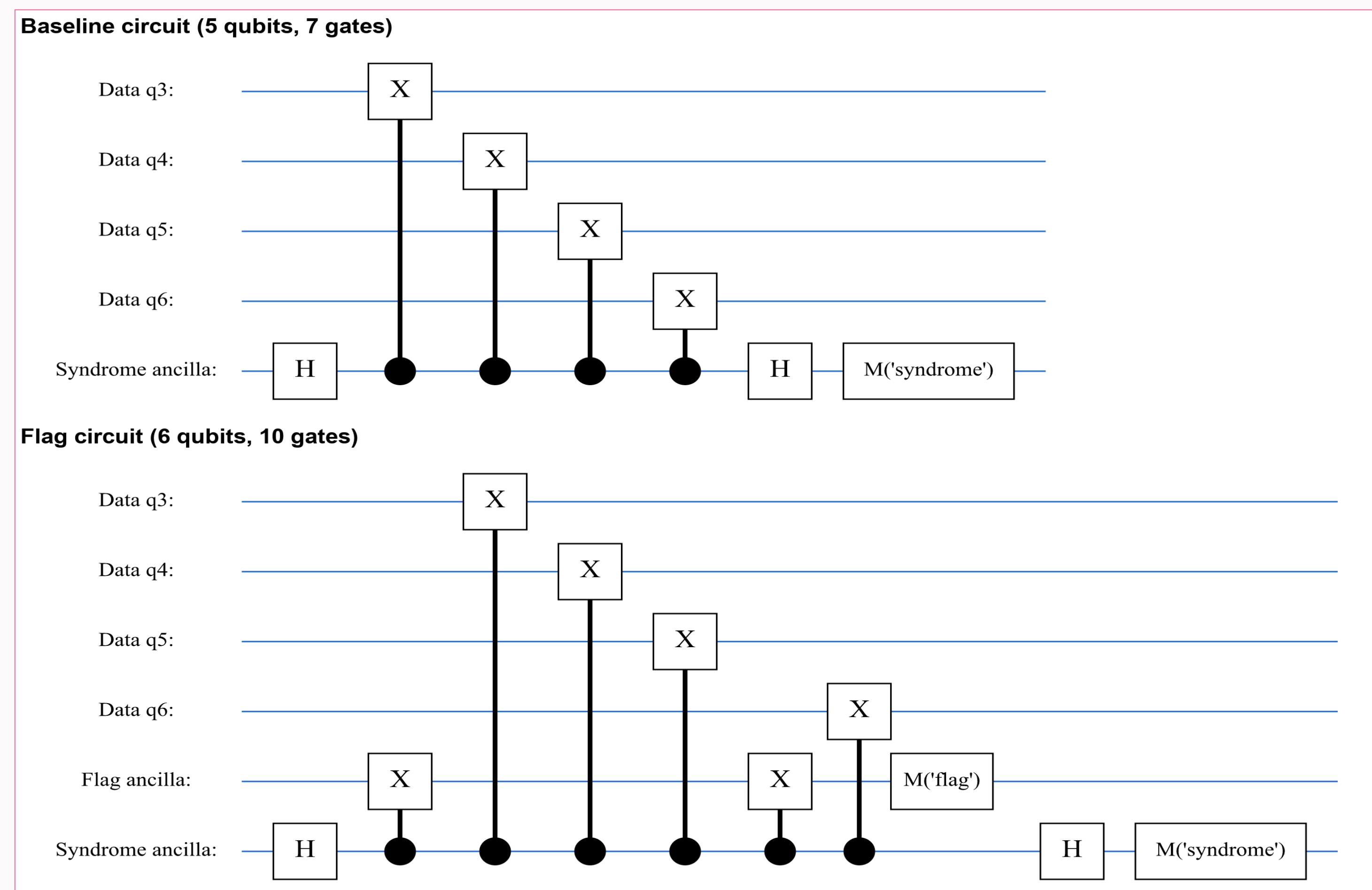


Figure 1: Baseline and flag-qubit circuits for the $X_3X_4X_5X_6$ stabilizer. Both circuits measure the same weight-4 X-stabilizer using an ancilla prepared in the $|+\rangle$ state and coupled to data qubits 3–6 via CNOTs.

- Two circuits were implemented for one weight-4 X-stabilizer of the Steane code ($X_3X_4X_5X_6$):
 - a standard baseline ancilla circuit,
 - a flag-qubit circuit in which a second ancilla opens and closes a detection “window” around the two riskiest CNOTs in the sequence.
- Verified both against a clean, fault-free state before injecting any faults
- Enumerated every gate location in each circuit and injected each of the three single-qubit Pauli faults (X, Y, Z) at every location, one fault at a time
 - 30 total tests on the baseline circuit, and 42 on the flag circuit.
- Propagated each injected fault forward through the remaining gates using a stabilizer frame model to determine its final effect on the data qubits and ancillas.
- Classified each fault as correctable, detected by flag, or not correctable, then compared failure rates and qubit/gate overhead between the two circuits.

RESULTS

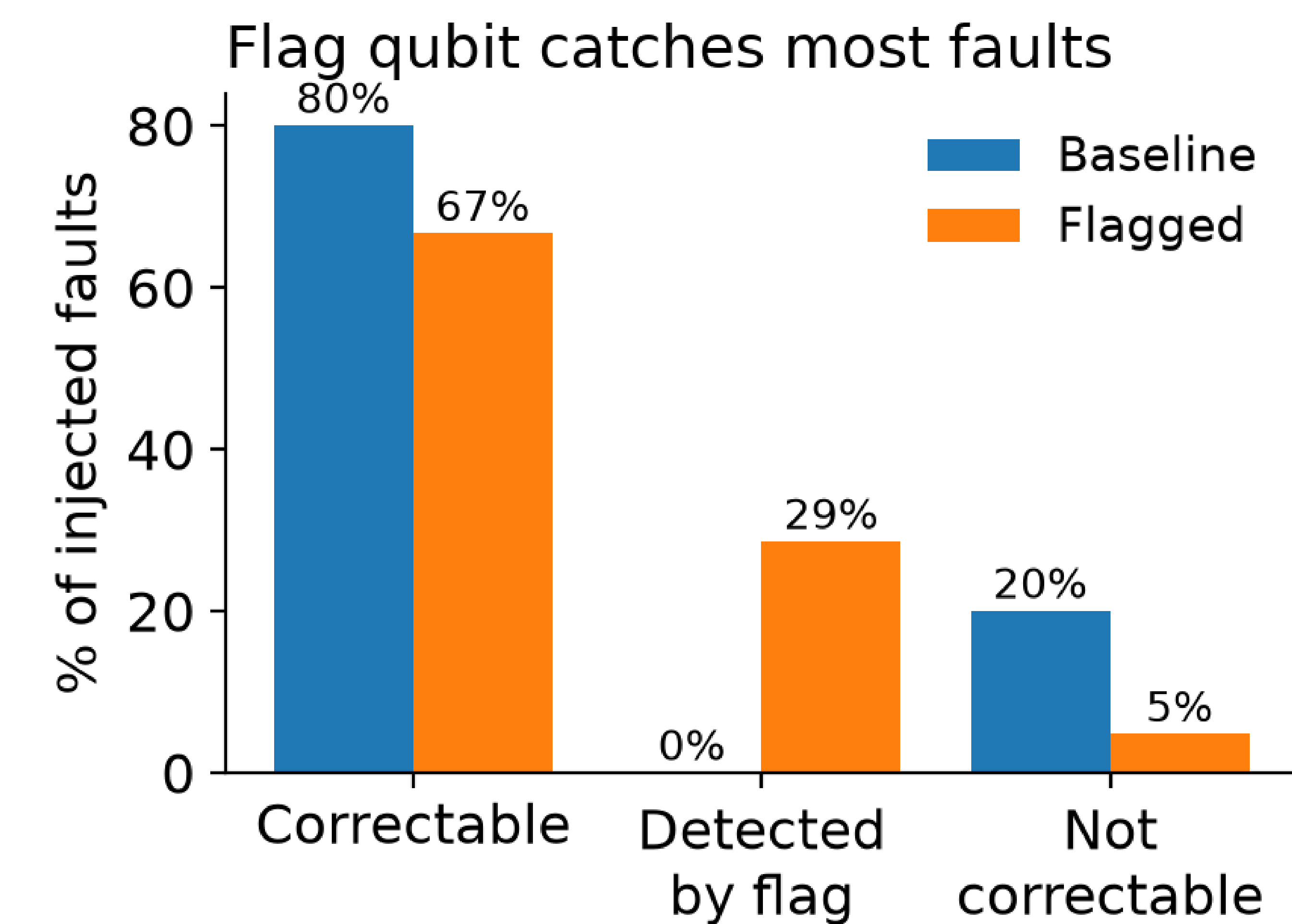


Figure 2: The flag qubit converts silent failures into detected events. In the baseline circuit, 20% of all injected faults produced an uncorrectable, undetected data error. In the flag circuit, the errors were almost entirely intercepted: 28.6% were caught by the flag ancilla, and the uncorrectable-and-undetected rate fell to 4.8%.

Circuit	Injected Faults	Correctable	Detected by Flag	Not Correctable	Qubits	Gates
Baseline	30	24 (80.0%)	0 (0.0%)	6 (20.0%)	5	7
Flag	42	28 (66.7%)	12 (28.6%)	2 (4.8%)	6	10

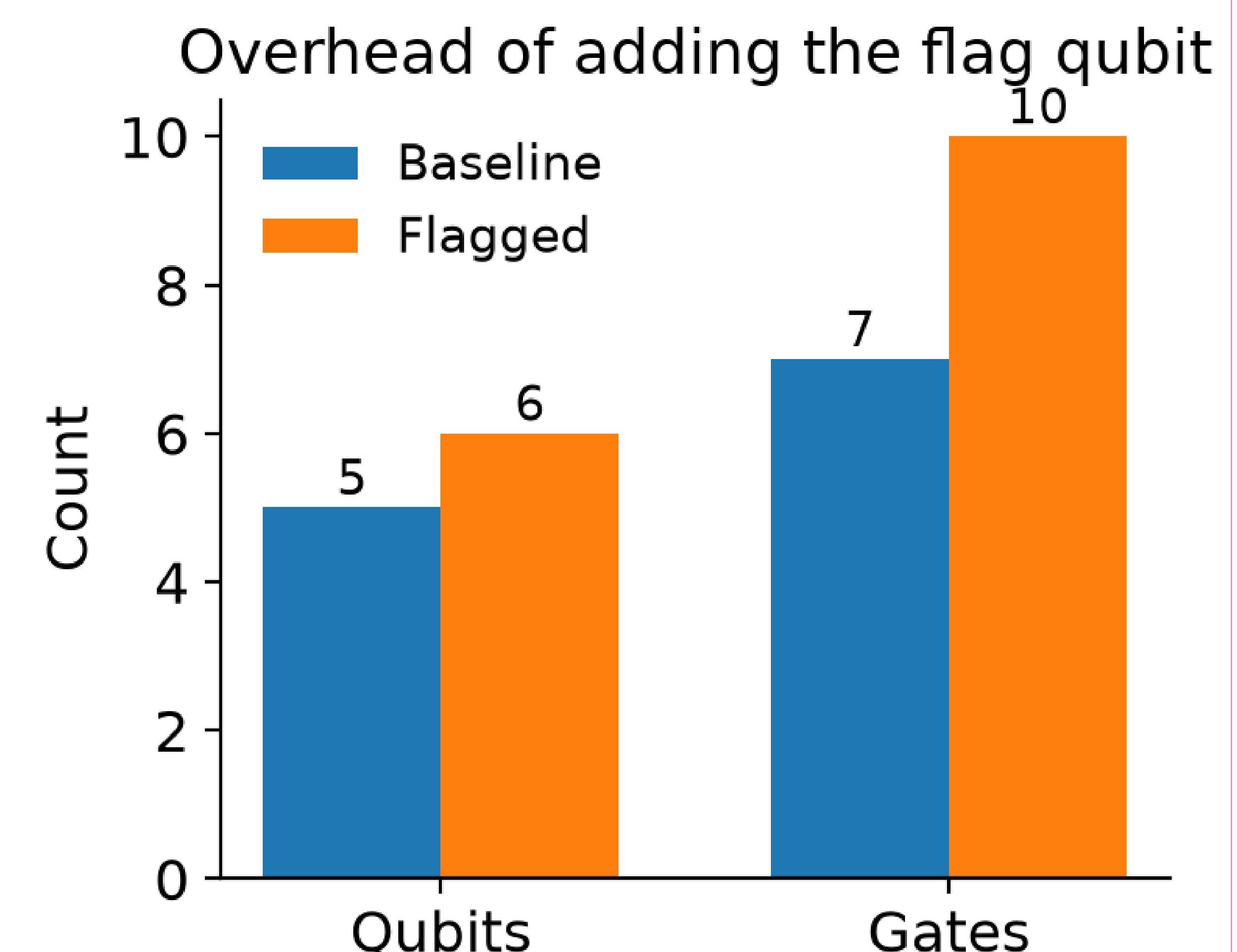


Figure 3: Fault detection costs one extra qubit and three extra gates. The baseline circuit uses 5 qubits and 7 gates; the flag circuit uses 6 qubits and 10 gates. One ancilla and three gates are enough to reduce the undetected-failure rate by more than 75%.

Table 1: Fault-outcome counts and circuit overhead for the baseline and flag-qubit stabilizer measurement circuits out of 30 and 42 injected single-qubit Pauli faults, respectively. The flag circuit converts most of the baseline's not-correctable faults into detected-by-flag outcomes, cutting the undetected-failure rate from 20.0% to 4.8%, at a cost of one additional qubit and three additional gates.

CONCLUSION

A single flag qubit reduced the rate of undetected, uncorrectable faults from 20% to 4.8%, at a cost of only one additional qubit and three additional gates. Most of the faults that the flag circuit “removed” from the uncorrectable category were made visible, moving into the detected-by-flag category where a decoder can act on them rather than silently corrupting the logical qubit.

This result is a concrete demonstration of what a fault-tolerant gadget is meant to do. Our flag qubit acted as an inspector gadget for the moment a syndrome-ancilla fault can spread across several data qubits. It converted a silent, uncorrectable failure into a flagged, actionable one, and at a fixed and modest cost, too. This is the same design logic behind the broader family of fault-tolerant gadgets, and it's what makes flag-qubit schemes a practical near-term building block for scalable fault tolerance.

REFERENCES and ACKNOWLEDGEMENTS

- ¹Devitt, Simon J., William J. Munro, and Kae Nemoto. “Quantum Error Correction for Beginners.” *Reports on Progress in Physics*, vol. 76, no. 7, 2013, article 076001.
- ²Reichardt, B. W. (2020). Fault-tolerant quantum error correction for Steane's seven-qubit color code with few or no extra qubits. *Quantum Science and Technology*, 6(1), 015007.
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